

# AEBN30 Lab Poster – Group 9

## Solar Thermal Lab Experiment 1

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### Efficiency of a Solar Heating System

#### Introduction:

This experiement analyses the efficiency of a solar thermal (ST) panel system. A ST system uses solar radiation to gain heat energy for water heating. A lab environment is utilized to collect the data, rather than using actual solar irradiance on the system. The aim is to understand how the efficiency of the system varies with the temperature difference between the solar collector and the surroundings.

#### Measurements and Instrumentation:

- **Irradiance:** The incident light on the collector surface is measured using a pyranometer (W/m<sup>2</sup>).
- **Temperatures:** Temperature sensors are placed at three points: water into the absorber, water out of the absorber, and water in the storage tank, plus ambient air temperature.
- **Data Logging:** All measurements are recorded automatically every 30 seconds by a data logger and software (Logger Pro).

#### Methodology:

A small-scale solar heating system is assembled with a black metal absorber plate housed in an insulated box with glazing. Water pipes run along the absorber, circulating water to a storage tank using a pump. In the laboratory, a halogen lamp simulates solar irradiance due to limited natural sunlight.

#### Data Analysis:

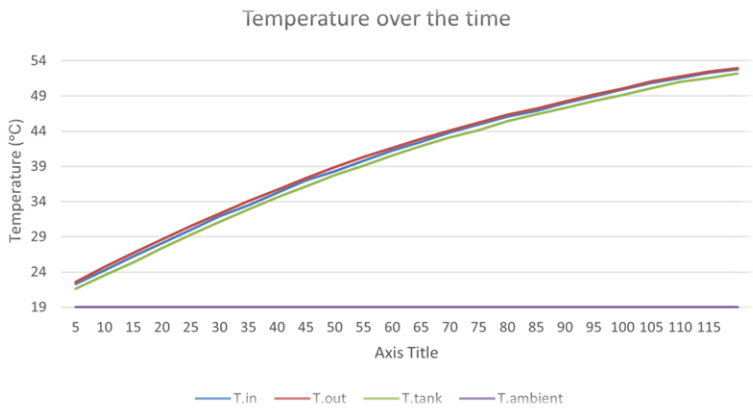
- Export collected time-series data and input into the provided Excel analysis sheet (template from course website).
- Discard initial readings and begin calculations after 5 minutes for steadier conditions.
- Compute **Power in** (*P<sub>in</sub>*) using measured irradiance and known collector area.
- Calculate the temperature difference ( $\Delta T$ ) between collector and ambient air for each interval.
- For each 5-minute interval, determine **Power out** (*P<sub>out</sub>*) by the heat delivered to the tank (from water temperature rise and flow/mass).
- Calculate instantaneous efficiency:  $\eta = P_{out}/P_{in}$ , and copy calculations for all time steps.
- Plot system temperature curves versus time, and chart efficiency as a function of  $\Delta T$ .

#### Specification:

|                   |                      |
|-------------------|----------------------|
| Irradiance        | 700 W/m <sup>2</sup> |
| Absorber area     | 0.12 m <sup>2</sup>  |
| Heated water mass | 2.7 kg               |

#### Formula:

$$P_{in} = G \cdot A \qquad \eta = \frac{P_{in}}{P_{out}}$$
$$P_{out} = \frac{m \cdot c_p \cdot \Delta T_{tank}}{\Delta t}$$
$$\Delta T' = \frac{(T_{in} + T_{out})}{2} - T_{ambient}$$



We can see that the system is heating up but over time the curve flattens out, as heat losses increases as the collector gets hotter.

#### Discussion:

The experiment shows the relationship between heat loss and the efficiency of the solar thermal system. The analysis tells us that the efficiency decreases as the temperature difference between the collector and ambient air increases.

- **What happens to the efficiency? Why?**

Efficiency decreases as  $\Delta T$  increases because higher the  $\Delta T$  makes greater conductive, convective and radiative heat loss from the collector to the surrounding which results in reduced *P<sub>out</sub>*.

- **If  $\Delta T=0$ , what happens?**  
There will be no heat loss, and the system runs at maximum efficiency.

- **If efficiency is 0, what happens?**  
It means that the system has reached thermal equilibrium and no energy is being transfer to the water.

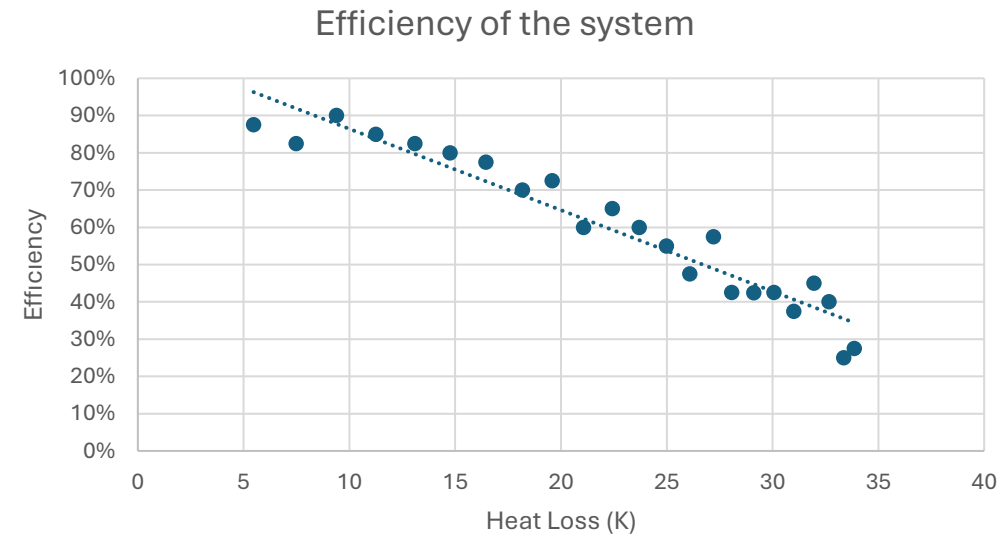
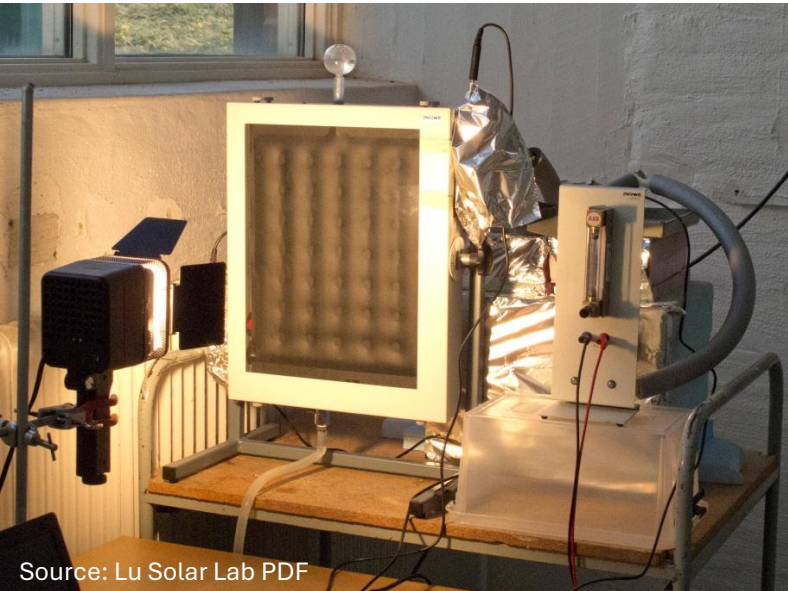
- **If you could reduce the heat losses from the collector, how would the efficiency curve be affected?**  
It will make the curve flatter, the system will run at higher efficiency.

#### Comments:

The measured efficiency represents the whole system, for a more accurate efficiency measurement, a steady state method using inlet/ outlet temperatures and precise flow rate would be required.

#### Conclusion:

To operate the solar thermal panel system, the heat losses must be minimized to have higher output temperate to heat the interior spaces via using a radiator.



As the collector gets hotter, there is rapid loss of heat from the collector over time to it surrounding, reducing the thermal energy (*P<sub>out</sub>*) delivered to water, thus lowering the efficiency of the system.